Project Executable Files

Date	12 July 2026
Team ID	
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	3 Marks

Project Executable Files

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

```
OptiCrop/
│
├── __pycache__/
│
├── recordings/
│
├── static/
│   ├── css/
│   ├── js/
│   └── images/
│
├── templates/
│   ├── index.html
│   ├── crop.html
│   ├── fertilizer.html
│   ├── disease.html
│   └── result.html
│
├── README.md
│
├── app.py
│
├── convert_recordings.py
│
├── crop_model.pkl
│
├── requirements.txt
│
└── test_app.py
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://smartagriculture-one.vercel.app/
Login Credentials (Demo)	Not Applicable (No login required)
Platform / Hosting Provider	Vercel
Repository Link	Documentation repository: https://github.com/MohanaSuryadevara/Smart-Agriculture Code files repository: https://github.com/Abhi-4363/smart_agriculture
Demo Video Link	https://drive.google.com/file/d/1p0kNiHHF7aImcYjtNiXX4dBxpo4wne9R/view?usp=sharing

Step 4: Run Instructions

- Clone or download the project from the GitHub repository.
- Install Python 3.10+ and all dependencies using pip install -r requirements.txt.
- Ensure the trained model.pkl file is present in the project directory.
- Run the Flask application using python app.py.
- Open the local URL shown in the terminal (or use the deployed Vercel link), enter the required soil and environmental parameters, and view the crop recommendation.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Recommendations are based only on the trained dataset and provided input parameters.	Future versions will integrate real-time weather and market data.
2	The prediction model is trained on a specific dataset and may require retraining for new crops or regional conditions.	The model can be updated periodically with new agricultural datasets.
3	Real-time weather and market price integration are not available in the current version.	Planned for future enhancement to provide more comprehensive recommendations.